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Unit – 1 : Session – 5 : SLO – 2 | Matrix Multiplication

■ I. Fill in the Blanks

1. columns — valid only if columns of the first matrix equal rows of the second.
2. rows of first — the result has dimensions (rows of first) $\times$ (columns of second).
3. corresponding — multiply each row element with the corresponding column element.
4. each element (repeated/dot-product) — a loop is required for each computation.
5. nested — use three nested loops (i, j, k).
6. zero — initialize result elements to 0.
7. dot — the innermost loop calculates the dot product.
8. accumulated (running sum) — $\text{sum} += a[i][k] * b[k][j]$.
9. result — store the computed value in the result matrix $\text{res}[i][j]$.
10. dimension mismatch (runtime/logical) error.
11. 2D arrays (contiguous memory) — matrix elements are stored in two-dimensional arrays.
12. use (computation) — initialize arrays before computing.
13. dimensions — declare the result with proper $r1 \times c2$ dimensions.
14. 2×4 — $(2 \times 3) \times (3 \times 4) \rightarrow 2 \times 4$.
15. integer (element-wise multiply-accumulate) arithmetic.